

DAT410 - Design of AI Systems

Group 69, Module 4		
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We hereby declare that we have both actively participated in solving every exercise. All solutions are entirely our own work, without having taken part in other solutions.

1. Reading and reflection

- Read and summarise (briefly) the papers [Machine learning techniques to diagnose breast cancer from image-processed nuclear features of fine needle aspirates](#) and [The Mythos of Model Interpretability Links to an external site](#). Your take-aways from these papers will be used in the implementation and discussion.

The machine learning techniques to diagnose breast cancer from image-processed nuclear features of fine needle aspirates (FNA) describes a computer system that diagnoses breast cancer from FNA using digital image analysis and machine learning. The system was developed with data from 569 initial patients (212 cancer, 357 benign) and was tested on 54 new patients. The system employed “snakes” (active contour models) to automatically identify cell nuclear boundaries from digital images. 10 nuclear features were measured namely: radius, perimeter, area, compactness, smoothness, concavity, concave points, symmetry, fractal dimension, and texture. Since multiple nuclei were analyzed, a single set of features per patient is calculated by 3 statistics for each of the 10 measurements; mean, worst (mean of 3 largest), and standard error. Classification was done using MSM-Tree (Multisurface Method-Tree) using linear programming to create separating planes. The best classifier used only 3 features (mean texture, worse area and worst smoothness) to achieve 97% cross-validation accuracy. The test result was ultimately 100% accuracy on 54 new samples (17 malignant, 36 benign, 1 atypical). Although simple, when a simple linear model is applied correctly, excellent performance can be achieved with only 3 features.

The 2nd paper examines the concept of “interpretability” in machine learning, arguing that the term is poorly defined and often used inconsistently in literature. There are 2 main arguments. One that is interpretability is not monolithic which means that different stakeholders want different things. Two that there are 2 main categories of interpretability:

- 1) Transparency (understanding the model itself): algorithmic transparency, decomposability, simulatability
- 2) Post-hoc explanations (explaining individual predictions): text explanations, visualizations, local approximations

Although linear models are simplistic, they are often over simplified. With high-dimensional or heavily engineered features, linear models lose their interpretability despite algorithmic transparency. Paradoxically, deep neural networks, though architecturally complex, may use more interpretable raw features while linear models may require more complex feature engineering, thus trade-offs exist when we talk about the model concerning its interpretability. Interpretability highly depends on context too, where the application is of main concern—medical diagnosis has different requirements than product recommendations.

Motivations for interpretability may also vary:

- Trust - will the model work in deployment?
- Causality - will the model understand relationships?
- Transferability - will it generalize?

- Informativeness - what can it teach us?
- Fair and ethical decision making - can it make sound decisions?

Ultimately researchers should specify what kind and for whom is interpretability for rather than claiming it generally – to be specific about goals rather.

2. Implementation

- The file [wdbc.pkl Download wdbc.pkl](#) contains a dataset of patients tested for breast cancer using a fine needle aspirate (FNA). Each patient is represented by statistics of 10 different features of multiple cell samples, as well as a diagnosis (malignant or benign). For definitions of these variables, see [Breast Cancer Dataset. Links to an external site.](#) The file [wdbc.pkl Download wdbc.pkl](#) has columns in the order: subject id, label, features. *Clarification: The three different features for each variable, with suffixes "_0", "_1", "_2" represent the mean, standard deviation and "worst" value over a set of samples, respectively. Please see the "Data description" Links to an external site. for a precise definition.*
- Identification of breast cancer from FNA is based on a number of factors which make tumor cells stand out among regular tissue [[Pfragner & Freshney, 2003 Links to an external site.](#)].
- [Eickhoff \(2016\) Links to an external site.](#) ([pdf here Download pdf here](#)) summarizes these as follows: "The size of cells tends to be homogeneous given a specific type of tissue. 1) The presence of significantly larger cells is evidence for the uncontrolled growth that is indicative of malignant tumors. 2) The shape of benign cells usually shows only limited variance, whereas malignant cells can develop arbitrary structures that do not conform with the general pattern of their surroundings. 3) The color of the cell nucleus should be identical for regular cells of the same type. Cancer cells often have significantly larger and darker nuclei that are more densely packed with DNA. 4) Regular cells show similar texture. Malignant tumors, on the other hand, can range from smooth surfaces to ragged or lumpy textures for neighbouring cells. 5) Finally, for healthy tissue, cell arrangement tends to be orderly, with regular distances between cells. Cancer cells can spread out or clutter almost arbitrarily.
- Implement and evaluate the following classifiers in the task of assigning the diagnosis *malignant* or *benign*:
 1. A rule-based classifier which follows the following form:
 - If *[cell size is abnormal]*:
 - or *[cell shape is abnormal]*
 - or *[cell texture is abnormal]*
 - or *[cell homogeneity is abnormal]*,
 - then: diagnosis is *malignant*,
 - otherwise: diagnosis is *benign*.
 - *There feature names in the Eickhoff summary may be slightly different—you must interpret what these rules should be in the context of the available data.*
 2. A random forest classifier using the sklearn framework applied to the features given in the supplied dataset.
 3. A classifier of your own design which attempts to trade off interpretability and classification performance. *Clarification: This can build on existing models but the choice/design should be focused on the mentioned trade-off.*

- For the first classifier, it is up to you to define the appropriate variables and the notion of abnormal, based on the supplied data. Motivate your choices. *There is not only one solution to this problem. The purpose is to translate medical insights into a model based on available data, and to provide a baseline for other classifiers.*
- Compare the classification performance of the three classifiers and comment on their interpretability, from your perspective. (Note that your interpretation may differ from that of a physician). Are there notable interactions between features?

Data Pre-Processing

Data pre-processing was conducted to ensure only relevant variables were used as input features for model training. Our objective is to learn the mapping function $f(X) = y$, where X denotes the feature matrix and y denotes the outcome variable.

For this dataset, the columns *id* and *malignant* were removed from the dataframe. The *id* column represents a unique patient identifier and does not contain any medical or biological information related to tumor characteristics. Including this variable can potentially introduce noise and create bias in the model, as it has no predictive value. Hence, it was removed to ensure only clinically relevant measurements were used for classification. The *malignant* column represents the diagnosis label (1 = malignant, 0 = benign), which is the target variable that the model is trained to predict. For a training set, this column would have to be removed as it would lead to data leakage if included. The model would be able to access the true outcome during training and artificially inflate performance metrics.

Classifier 1: Rule-Based Classifier

This classifier was constructed by directly translating medical insights from Eickhoff (2016) into explicit rules and logic based on the measurable features below:

Medical Characteristic	Corresponding Features
Cell size	Radius, Area, Perimeter
Cell shape	Compactness, Symmetry
Cell texture	Texture, Smoothness
Cell homogeneity	Concavity, Concave Points

Abnormality thresholds were determined using training data statistics. Specifically, thresholds were calculated as midpoints between the 75th percentile of benign samples and the 25th percentile of malignant samples. This approach aimed to separate the distributions while remaining data-driven.

The final rule uses OR-based logic to reflect the assumption that any strongly abnormal cellular property may indicate malignancy, flagging it down for further review. The model works as follows:

IF (Cell size is abnormally large) **OR** (Cell shape is abnormal) **OR** (cell texture is abnormal) **OR** (Cell homogeneity is abnormal) **THEN** malignant **ELSE** benign

The final results for the Rule-Based Classifier are as follows:

- Training Accuracy: 41.10%
- Test Accuracy: 39.47%
- Confusion Matrix:
 - $\begin{bmatrix} 3 & 69 \\ 0 & 42 \end{bmatrix}$
- Performance metrics for malignant class:
 - Precision: 0.378
 - Recall (Sensitivity): 1.000
 - F1-score: 0.549

The classifier achieved perfect recall, where no malignant cases were missed. However, the model severely over-predicts malignancy, leading to an extremely high false positive rate. Only 3 of 72 benign cases were correctly classified. The overall accuracy is therefore very low. The primary issue lies in the OR-based structure. Malignancy affects multiple features simultaneously, and even small deviations in benign samples trigger malignant predictions. The model lacks the ability to combine evidence in a more nuanced way.

The data is very interpretable, as the decision rules are fully transparent (decided on OR rules) and easily explainable. Every prediction can be traced back to specific biological criteria. However, interpretability alone does not compensate for poor predictive performance.

Classifier 2: Random Forest Classifier

Random Forest was selected for its ability to model complex nonlinear interactions and high-dimensional data. The classifier was trained using:

- 100 decision trees
- Maximum depth = 10
- Minimum samples split = 5
- Minimum samples per leaf = 2
- Random state = 42

After training the Random Forest Classifier using all 30 available tumor features, feature importance scores were extracted using the impurity-based importance measure provided by the RandomForestClassifier implementation in scikit-learn (sklearn). These values are automatically computed by scikit-learn during model training based on how strongly the variable contributes to the classification process.

During training, each tree recursively splits the data using different features in order to reduce class impurity. When a feature is used to split the data and results in a significant reduction in impurity, it would contribute positively to the feature's importance score. The total importance score is then calculated as the sum of all impurity reductions across all trees, normalized so that the importance scores sum to one. The features were then ranked in descending order according to their performance scores, and the top most important features were purely selected based on the ranking.

From the classifier, the most important features were:

1. area_2
2. concave points_2
3. radius_2
4. concave points_0
5. perimeter_2
6. perimeter_0
7. radius_0
8. concavity_0
9. area_0
10. concavity_2

This shows that size-related features (area_2, radius_2, perimeter_2) and boundary-related features (concave points_2, concave points_0) are the most influential predictors within this model. Another notable pattern is that the “worst” values (with the suffix _2) are ranked highly, indicating that extreme tumor characteristics are particularly informative for distinguishing malignant tumors.

The final results for the Random Forest Classifier are as follows:

- Training Accuracy: 99.34%
- Test Accuracy: 97.37%
- ROC-AUC: 0.9967
- Confusion Matrix:
 - $\begin{bmatrix} 72 & 0 \\ 3 & 39 \end{bmatrix}$
- Performance metrics for malignant class:
 - Precision: 1.000
 - Recall (Sensitivity): 0.9286
 - F1-score: 0.963

In this case, only three malignant cases were misclassified. The Random Forest achieves excellent predictive performance and near-perfect ROC-AUC. It successfully balances false positives and false negatives. However, interpretability is limited due to the use of complex nonlinear interactions of the random forest model. While feature importance provides further insight, individual predictions cannot be easily explained. In clinical settings, this may reduce trust and complicate any regulatory approval processes.

Classifier 3: Decision Tree Classifier

To balance interpretability and performance, a constrained decision tree was developed.

- Maximum depth = 4
- Limited to 10 medically meaningful features — this selection was made on medical criteria as described in Eickhoff (2016) where features map to 4 diagnostic indicators: cell size (radius, perimeter, area), cell shape (symmetry, compactness), texture, and cell homogeneity (concavity, concave points)
- Focused primarily on worst values ($_2$) — the most abnormal cells in a sample may carry greater diagnostic significance than the average measurements, consistent with medical intuition around cancer cell behaviour. A single severely abnormal cell is more meaningful than a collection of normal ones.

This structure ensures that the decision process (benign or malignant) remains understandable and auditable without sacrificing competitive performance.

The final results for the Decision Tree Classifier are as follows:

- Training Accuracy: 95.82%
- Test Accuracy: 91.23%
- ROC-AUC: 0.9539
- Confusion Matrix:

	Predicted Benign	Predicted Malignant
Actual Benign	72	0
Actual Malignant	10	32
- Performance metrics for malignant class:
 - Precision: 1.000
 - Recall (Sensitivity): 0.7619
 - F1-score: 0.8649

The model produced 0 false positives, where every case predicted as malignant was confirmed malignant, yielding perfect precision. However, this should be interpreted with caution, as it may partly reflect the relatively small test size rather than a guaranteed property of the model. More importantly, 10 malignant cases were misclassified as benign (false negatives), meaning patients with cancer would have been incorrectly told they were clear. In medical diagnostics, false negatives carry significant risk and recall is therefore a more critical metric to look at. At 76%, this is a meaningful limitation that a clinician would need to weigh against the model's transparency.

We can reproduce the full decision tree structure as below:

```
|--- perimeter_2 <= 112.80
| |--- concave points_2 <= 0.16
| | |--- concave points_2 <= 0.14
| | | |--- radius_0 <= 14.86
| | | |--- class: 0
```



```

| | | |--- radius_0 > 14.86
| | | |--- class: 0
| | |--- concave points_2 > 0.14
| | | |--- texture_2 <= 24.44
| | | |--- class: 0
| | | |--- texture_2 > 24.44
| | | |--- class: 0
| | |--- concave points_2 > 0.16
| | |--- class: 1
|--- perimeter_2 > 112.80
| |--- concave points_2 <= 0.14
| | |--- texture_2 <= 25.61
| | | |--- class: 0
| | | |--- texture_2 > 25.61
| | | |--- class: 1
| |--- concave points_2 > 0.14
| |--- class: 1
(Depth: 4, Leaves: 8)

```

The tree can be read as a clinical flowchart with a small number of questions, each grounded in a measurable feature:

Step 1: Is the worst perimeter large?

The root split on `perimeter_2 <= 112.80` separates tumours by overall cell size. A worst perimeter above 112.80 immediately flags a case as high suspicion. This aligns with the well-established medical criterion that malignant tumours tend to produce larger, more irregular cells.

Step 2: Are the cell boundaries highly irregular?

Regardless of which branch is taken at step 1, the next key question is `concave points_2`: the number of severe concave indentations on the cell boundary. This feature appears at multiple levels of the tree and is effectively the strongest secondary discriminator. A high value (>0.16 for smaller tumours, >0.14 for larger ones) pushes the prediction strongly toward malignant. Irregular, lumpy boundaries are a hallmark of uncontrolled cell growth.

Step 3: Does texture confirm the suspicion?

For the borderline cases where perimeter is large but concave points are low, the model introduces `texture_2` as a tiebreaker. A worst texture above roughly 25 tips the prediction towards malignant. Texture here measures variation in cell surface intensity, and higher variation indicates abnormality.

Step 4: Final boundary cases

Within smaller perimeter branches, `radius_0` appears at the deepest level but both of its leaf outcomes are benign (class 0), suggesting that once a tumour is small and has low concave points, mean radius does not add discriminating power, it simply provides a structural split the tree learned without practical effect.

Taken together, the tree encodes a clinically coherent logic: Large cells with irregular, deeply concave boundaries and rough surface textures are classified as malignant. This mirrors how a pathologist would reason about a cytology slide, and the entire decision process can be

reproduced with just 3 feature measurements in most cases.

Interactions between Features

Strong correlations were observed between

Size

- radius_2 ↔ perimeter_2: 0.994
- radius_0 ↔ perimeter_0: 0.998
- area_2 ↔ radius_2: 0.983

Radius, perimeter and area are all near-perfectly correlated ($r > 0.98$), as they all describe geometrical scale.

Concavity

- concavity_0 ↔ concave points_0: 0.921
- concavity_2 ↔ concave points_2: 0.849

Concavity and concave points ($r = 0.92$) consistently co-occur in malignant samples.

In addition, worst values consistently show higher predictive power than mean values. Malignancy affects multiple cellular properties simultaneously:

- Increased cell size
- Irregular boundaries
- Higher concavity
- Increased variation in structure

The high correlations suggest coordinated biological changes rather than isolated abnormalities.

Evaluation and Conclusion

In summary, the interpretable decision tree performs significantly better than the rule-based classifier and maintains a reasonable accuracy relative to the RF model, despite using far fewer features and is limited by depth. This trade-off however is a lower recall of 76%, which is a genuine clinical limitation. However, its key advantage lies in the fact that every prediction can be traced through a short auditable decision path that a clinician can review, challenge and validate. This satisfies explainability requirements that a black-box model like the Random Forest cannot, and makes the model a viable candidate for decision-support in a clinical setting where trust and transparency are as important as raw performance.

However, in the context of breast cancer diagnosis, recall becomes a decisive factor. The most critical error is a false negative, where a malignant tumor is incorrectly classified as benign. Such an error may result in delayed treatment and allow the disease to progress undetected. In contrast, a false positive may lead to additional medical examinations, but it does not overlook a

potentially life-threatening condition. For this reason, minimizing false negatives is of primary importance in medical classification tasks.

This clinical priority is reflected in the model results. The Random Forest classifier produced 3 false negatives out of 42 malignant cases, achieving a recall of 93%, while maintaining high overall accuracy. In comparison, the Decision Tree classifier produced 10 false negatives, missing a substantially larger proportion of malignant tumors. Although the Decision Tree model is more transparent and easier to interpret, the increased number of missed malignant cases represents a meaningful clinical limitation.

Therefore, the choice between models depends on the intended deployment context. If maximizing detection of malignant tumors is the primary objective—such as in screening applications—the Random Forest model is preferable due to its superior recall and overall predictive performance. However, if explainability, auditability, and clinician trust are critical requirements—particularly in regulated medical environments—the interpretable decision tree remains a viable and defensible alternative, despite its lower recall.

Ultimately, this comparison illustrates the fundamental trade-off between predictive performance and interpretability in medical AI systems, highlighting the need to carefully balance clinical safety with transparency and trust in real-world deployment.

3. Discussion

- Interpretability of systems, models and predictions is important in many applications. Healthcare is one of these. Discuss the meaning of interpretability in the context of this week's assignment, its potential benefits and drawbacks, and how interpretability may be defined. Be as specific as you can and use mathematical notation where appropriate. Take inspiration from e.g., "The Mythos of Model Interpretability".

Since this article is about healthcare, predicting a benign when malignant can negatively impact one's treatment and lifespan. In this case a false negative (predicting benign when malignant) could prevent life-saving treatment, while a false positive (predicting malignant when benign) subjects patients to unnecessary biopsies, anxiety, and potential complications.

Drawing from Lipton's taxonomy, interpretability in breast cancer diagnosis can be understood through multiple lenses:

- 1) Transparency
 - a) Simulatability (understanding how the model works) — we can think of this as “can a clinician mentally trace through the model's decision process”
 - b) Decomposability (can each component be understood intuitively?) — features such as “mean feature” and “worst smoothness” have clear morphological interpretations and parameters representing the association strength between area and malignancy but these weights are fragile because if we add a correlated feature (e.g. “worst perimeter”), the weights can flip signs without changing predictions.
 - c) Algorithmic transparency (do we understand the learning algorithm's properties?) — linear models guarantee convergence, unique solutions and convex optimization while neural networks have non-convex optimization, no convergence guarantees, sensitive to initialization. And this is important for deployment as retraining on new data might not guarantee stable results
- 2) Explaining individual predictions (Post-hoc interpretability)
 - a) Explanation by example: “this tumour is classified as malignant because it resembles these other malignant cases” — Could show k-nearest neighbours in the learn feature space which aligns with how pathologist reason by analogy to previous cases but it doesn't explain why these features matter rather they provide intuitive support
 - b) Feature importance: “the model's decisions were most influenced by $\text{worst_area} = 1200$ and $\text{worst_smoothness} = 0.15$ ” — For linear models they directly read coefficients but for more complex models we could use SHAP values. However, a critical limitation is that these are only local explanations, moving the values to a slightly different case might produce completely different feature importances.
 - c) Natural language explanation: “the tumour shows irregularly shaped nuclei with high size variation” — mapping model's internals to clinically meaningful language could be useful as they can be generated by a separate model trained to explain the predictor. However the explanation model might produce plausible

but incorrect justifications, meaning the model might learn to generate text that sounds medically plausible without actually describing what the predictor model does. This results in clinicians who might trust this explanation and miss that the model is actually making decisions based on something entirely different. (unintentionally deceiving model)

3) Benefits of interpretability in this application

- a) Trust and clinical adoption — if physicians can't understand why a model makes predictions, they are more unlikely to trust it, especially when it contradicts their judgment. A transparent model therefore facilitates calibrated trust and error detection. Physicians are able to trust the model in cases where it's reliable and maintain skepticism where it's uncertain. Where error is seen, it can be on the physician to make judgement in order to subject the patient to additional tests. E.g. the model classified based on `worst_area`, but the physician can see extensive necrosis then the physician would order additional tests.
 - b) Scientific discovery — the Wolberg study found that texture, area and smoothness were the most predictive and this can generate hypotheses about cancer biology (think why do these features correlate with malignancy?) and could guide future research into cellular mechanics. However, correlation $\neq$ causation as a model learns associations not mechanisms on how biology works. The why remains unknown.
 - c) Regulatory and legal compliance — medical AI systems may face regulatory requirements for explainability where patients may be entitled to understand automated medical decisions. This means that in practice if a ML model makes a decision that significantly affects you (e.g. denying a loan, rejecting a job application or diagnosing a medical condition) you have the right to know that the automated decision-making was used, understand the logic behind the decision, contest the decision (if necessary) & request human review and receive meaningful information about how the algorithm works. However, what counts as explanation may vary e.g. the european GDPR is deliberately vague about what constitutes as an adequate explanation which has sparked major debates. This is because a maximalist vs a minimalist interpretation could be completely different. Furthermore there are technical challenges to this explanation as models are vastly different. Simple linear models with lesser features will be easier to explain as compared to deep neural networks (DNN) which may be more accurate but are harder to explain. This poses the question "Are we required to use less accurate models to comply with the law?"
 - d) Debugging and explanation — interpretability helps identify problematic model behaviour because if the model learns "orange objects" = cancer from biased training data then we can catch this spurious correlation. Additionally, if deployed in a new hospital with different staining protocols, we can diagnose why performance degrades with datasets shifts.
- ### 4) Drawbacks and limitation of prioritizing interpretability
- a) Accuracy-interpretability tradeoff — the Wolberg study achieved 97% accuracy with just 3 features but what if a DNN using all 30 features (plus learned

representations) achieves 99.5% accuracy? Is the cost of the 2.5% of improvement (in order to save hundreds of lives annually at scale) more than or less than transparency? Lipton also warns “Some arguments against black-box algorithms appear to preclude any model that could match or surpass human abilities on complex tasks.”

- b) Misleading interpretations from post-hoc explanations — consider 2 models with identical predictions:

Model A: $y^{\wedge} = \text{sign}(0.8 \cdot \text{area} + 0.2 \cdot \text{texture})$

Model B: $y^{\wedge} = \text{sign}(0.2 \cdot \text{area} + 0.8 \cdot \text{texture})$

If area and texture are highly correlated, these produce similar predictions but radically different “explanations” of what matters.

- c) The illusion of interpretability — Using a linear model with 3 dimensions makes it simple to explain but what if we use 30 dimensions. Is it truly interpretable then?

Lets say a linear model has 30 dimensions then we can write,

$$f(x) = \sum_{i=1}^{30} w_i x_i - b$$

Can a human then actually reason how 30 weighted features combine? The cognitive load exceeds that of human capacity, then we can say that this is only algorithmically transparent but not simulatable. Furthermore, the “worst” features (mean of 3 largest values) themselves involve complex preprocessing which would contend with whether or not this is interpretable.

- d) Feature engineering hides complexity — the 30 features involved sophisticated image processing that used the fractal dimension with coastline approximation, concavity computed from chords between non-adjacent snake points and the “snake” algorithm with energy minimization. However a physician can’t inspect a cell nucleus and mentally compute all these. The “interpretability” is therefore pushed into opaque preprocessing rather than the model itself.